

Towards Hierarchical Predictive Coding Predictors for Joint Embedding World Models

Abstract

Recent advances in Joint Embedding Predictive Architectures (JEPA) demonstrate that predicting latent representations rather than high-dimensional pixels enables scalable self-supervised representation learning. State-of-the-art frameworks—including I-JEPA, V-JEPA, and LeWorldModel—achieve highly performant downstream representation quality by optimization in latent space, effectively bypassing the computational overhead and trivial token-matching pitfalls of pixel-level reconstruction. Despite these algorithmic improvements, modern world models continue to rely heavily on static feedforward predictors optimized via global backpropagation. Conversely, biological cortical architectures execute perception through iterative hierarchical inference. In these systems, predictive representations propagate top-down, local prediction errors flow bottom-up, and weights undergo optimization via localized synaptic plasticity. This proposal investigates the integration of predictive coding as an alternative, biologically constrained inference mechanism within JEPA-style world models. Rather than altering the underlying encoder or modifying the objective functions, we propose replacing the standard feedforward predictor with a **Hierarchical Predictive Coding Predictor**. This module executes iterative latent-state optimization prior to generating the final target embedding. Additionally, inspired by cortical neuromodulatory mechanisms, we introduce an adaptive precision modulation framework to dynamically scale prediction-error propagation during inference. We evaluate this architecture on the task of image patch prediction within the I-JEPA framework. This setup provides a controlled baseline to directly isolate and compare the representational capacity of hierarchical predictive coding against conventional feedforward latent predictors.

1. Motivation & Background

The mammalian cortex processes sensory inputs as an active inference engine rather than a passive feedforward pipeline. Under the predictive coding paradigm, perceptual states emerge through the iterative minimization of localized prediction errors exchanged continuously between adjacent cortical layers. While modern self-supervised world models similarly focus on latent-space prediction, they collapse this dynamical inference process into a single forward pass through a deep neural network. This architectural gap motivates our primary research inquiry:

Can iterative predictive coding across a structured, hierarchical latent space yield richer, more robust, and more semantically organized representations than static feedforward prediction? Importantly, this work does not seek to replace existing

joint-embedding frameworks, but rather to redesign the inference topology that governs latent-state generation.

2. Research Hypothesis & Core Objectives

Hypothesis

Replacing the feedforward predictor in a Joint Embedding Predictive Architecture with a Hierarchical Predictive Coding Predictor improves the downstream quality and robustness of learned latent representations. This improvement is achieved by enabling the iterative refinement of multi-level latent states using only local message passing and localized synaptic updates.

Core Research Questions

1. **Inference Efficacy:** Can predictive coding frameworks serve as a stable, converging inference mechanism for multi-layer latent-state target prediction?
2. **Representational Topography:** Does iterative latent-state optimization generate superior semantic clustering and lower-dimensional manifold stability compared to feedforward alternatives?
3. **Credit Assignment:** To what extent can localized Hebbian plasticity variants optimize predictive world models without relying on end-to-end global backpropagation?
4. **Dynamic Scaling:** Does the introduction of an adaptive precision modulation layer improve the numerical stability and convergence rate of predictive coding networks during deep inference?

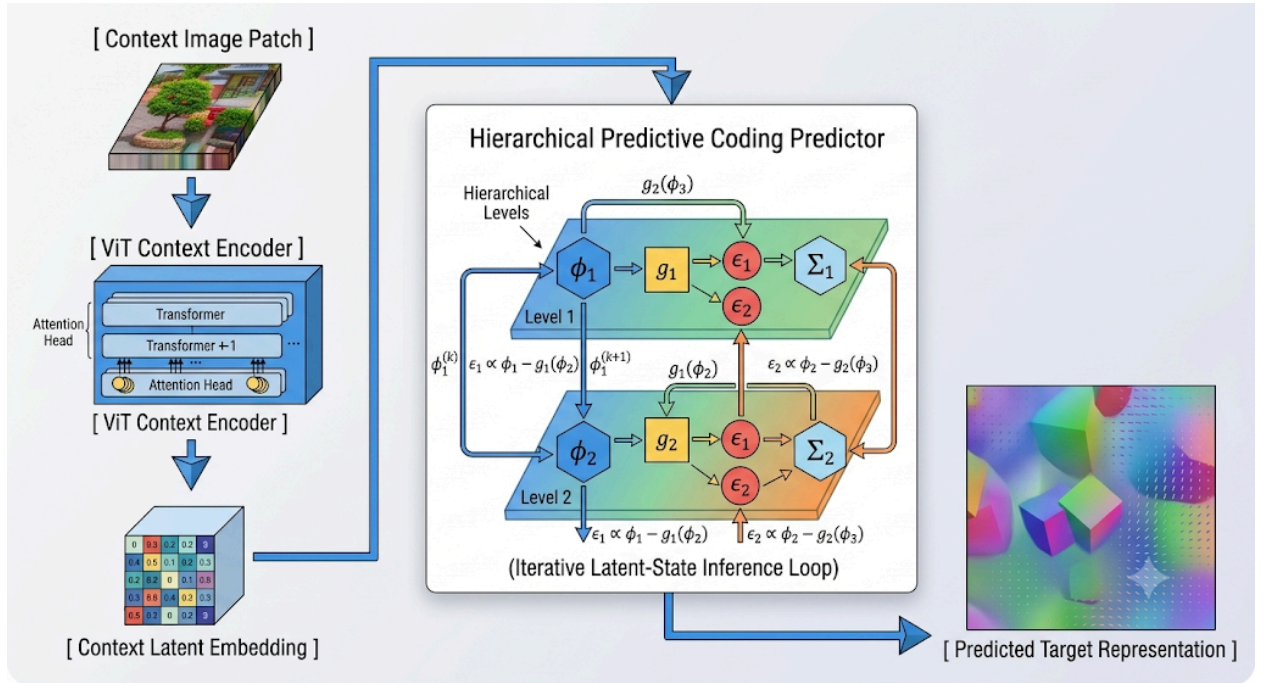
3. Proposed Architectural Framework and Predictive Coding Integration

The proposed framework preserves the established Vision Transformer (ViT) context encoder topology utilized in I-JEPA to provide a controlled baseline. We modify exclusively the predictor pipeline, replacing the conventional feedforward MLP with a dynamical, multi-level hierarchical inference system designed to isolate and measure the effects of predictive coding on representation quality.

3.1 Overview and System Flow

The overall flow operates across distinct computational pipelines (Figure 3.1). The standard encoder path processes input data to generate context-rich embeddings:

1. **[Context Image Patch]** $\rightarrow$ **[ViT Context Encoder]**: Input image patches are transformed into high-dimensional context latent embeddings through a standard frozen or pre-trained ViT.
2. **[Context Latent Embedding]**: This embedding serves as the foundational, non-predictive representation that drives the downstream predictor inference.
3. **[Hierarchical Predictive Coding Predictor]** $\rightarrow$ **[Predicted Target Representation]**: The context embedding is processed through our dynamic HPC-Predictor (described in Section 3.2), which executes an iterative latent-state optimization loop before outputting the final target latent.



This configuration allows direct comparison with the baseline (e.g., standard I-JEPA), where the same target representation is generated via a static forward pass through a multi-layer perceptron. The target branch (not visualized in detail but implied by the JEPA objective) remains an identical, parallel momentum-updated ViT pipeline that provides the stable target latent embeddings.

3.2 HPC-Predictor: Internal Structure and Inference Dynamics

The core innovation, detailed in the highlighted section of Figure 3.1, is the Hierarchical Predictive Coding Predictor. Unlike a feedforward net, this module is an open computational loop designed for deep hierarchical inference. It is structured into multiple levels (Level 1, Level 2, up to Level N), where each level progressively encodes more abstract and spatially stable latent representations.

Within each hierarchy i , we integrate four essential neural computational units:

- **Latent State Units (ϕ_i):** Dynamic representations that optimize over time, $\phi_i^{(k)} \rightarrow \phi_i^{(k+1)}$, until convergence.
- **Prediction Units (g_i):** Non-linear prediction functions (generative tiers) that project top-down hypotheses from the level above, $g_i(\phi_{i+1})$.
- **Prediction Error Units (ϵ_i):** Calculators of local mismatch, projecting bottom-up error gradients $\epsilon_i \propto \phi_i - g_i(\phi_{i+1})$ to update latent states at higher levels.

This specific internal topology enables the network to perform iterative inference—refining its beliefs across multiple hierarchies—prior to generating the final predicted target embedding used in the JEPA loss.

4. Mathematical Formulation

4.1 Hierarchical Inference Mechanics

Rather than computing the target representation in a single step, the predictor is formulated as a multi-tier hierarchical inference system. Each layer in the hierarchy maps to a distinct layer of representation depth, embedding four localized computational units: **Latent States (ϕ)**, **Predictions (g)**, **Prediction Errors (ϵ)**, and **Precisions (Σ)**. Top-down predictions flow from higher-level latent states to configure expectations for lower layers, while lower levels calculate and project prediction errors upward. The iterative state update for layer i is governed by:

$$\epsilon_i = \frac{\phi_i - g_i(\phi_{i+1})}{\Sigma_i}$$

Where:

- ϕ_i represents the current latent state vector at hierarchical tier i .
- $g_i(\cdot)$ denotes the top-down non-linear prediction function parameterized by weights W_i .
- Σ_i represents the uncertainty covariance or precision matrix associated with the layer. Latent states evolve dynamically through gradient descent on the total network free energy, driven exclusively by local prediction errors exchanged with adjacent layers until convergence is reached.

4.2 Localized Synaptic Plasticity

Once the fast latent-state inference loop reaches equilibrium, the slow synaptic parameters are updated via local Hebbian learning rules rather than global error backpropagation. The weight updates adhere to the generalized formulation:

$$\Delta W \propto (\text{Prediction Error}) \times (\text{Presynaptic Activity})$$

This formulation decouples the computational pipeline into two distinct biological timescales:

1. **Fast Latent Inference:** Iterative adjustment of state units (ϕ) to minimize localized prediction errors.
2. **Slow Synaptic Adaptation:** Weight updates (W) via Hebbian mechanics to optimize the predictive capacity of the generative tiers.

5. Experimental Protocol & Evaluation Metrics

Initial Downstream Task

The primary validation framework evaluates image patch prediction under the I-JEPA training paradigm. Restricting the initial scope to static images isolates the spatial predictive properties of the model from temporal modeling dynamics.

Evaluation Suite

To determine if iterative predictive coding yields higher-quality latent structures than feedforward pipelines, we implement a four-part evaluation protocol:

- **Linear Probing:** Training a frozen-weight linear classifier on the learned embeddings across standard benchmarks (e.g., ImageNet-1k) to assess linear separability.
- **k-NN Classification:** Evaluating non-parametric clustering properties of the latent space.
- **Manifold Topology Analysis:** Measuring the intrinsic dimensionality and smoothness of the latent space to quantify semantic abstraction.
- **Perturbation Robustness:** Measuring prediction stability and drift under targeted input corruptions, such as Gaussian noise, occlusions, and adversarial attacks.

Controlled Baseline Comparisons

The architecture will be evaluated directly against standard self-supervised baselines under identical constraints (same ViT backbones, pre-training datasets, and optimization budgets):

Baseline Model	Predictor Type	Optimization Type
Masked Autoencoder (MAE)	Feedforward (Pixel-level)	Global Backpropagation
I-JEPA	Feedforward (Latent-space)	Global Backpropagation
LeWorldModel	Feedforward (Recurrent Latent)	Global Backpropagation

Baseline Model	Predictor Type	Optimization Type
Ours (HPC-Predictor)	Hierarchical Predictive Coding	Localized Hebbian Plasticity

6. Expected Contributions & Future Outlook

Technical Contributions

- Design of a Hierarchical Predictive Coding Predictor adaptable to joint-embedding architectures.
- A bi-level optimization scheme separating fast iterative state inference from slow Hebbian parameter updates.
- A localized, neuromodulatory precision gating framework for stabilizing error propagation in deep architectures.
- Empirical benchmarks testing the representational efficiency of predictive coding against feedforward predictors in self-supervised learning.

Open Research Inquiries

While the theoretical foundations are mature, several engineering challenges will be addressed during the implementation phase:

- **Inference Latency:** Mitigating the computational overhead associated with running multi-step iterative loops inside every forward pass.
- **Asynchronous Execution:** Scaling localized Hebbian updates efficiently within modern, highly parallelized accelerator hardware (GPUs/TPUs).
- **Hyperparameter Stability:** Analyzing the convergence boundaries and numerical stability of deep hierarchical loops when integrated with high-capacity Transformer backbones.

Literature Survey

World Models & Joint Embedding Predictive Architectures

- **A Path Towards Autonomous Machine Intelligence** — Yann LeCun
<https://openreview.net/forum?id=BZ5a1r-kVsf>
- **Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture (I-JEPA)** — Assran et al. <https://arxiv.org/abs/2301.08243>

- **V-JEPA: Self-Supervised Learning of Video Representations by Predicting Masked Embeddings** — Bardes et al. <https://arxiv.org/abs/2404.08471>
- **V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning** — Meta FAIR <https://arxiv.org/abs/2506.09985>
- **LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels** — LeCun et al. <https://arxiv.org/abs/2603.19312>
- **LeJEPA: End-to-End Joint-Embedding Predictive Architectures with Sketched Gaussian Latent Regularization** — Balestriero et al. <https://arxiv.org/abs/2511.08544>

Predictive Coding & Free Energy Principle

- **Predictive Coding in the Visual Cortex: A Functional Interpretation of Some Extra-Classical Receptive Field Effects** — Rao & Ballard (1999) <https://homes.cs.washington.edu/~rao/Rao-Ballard-NN-1999.pdf>
- **Predictive Coding under the Free-Energy Principle** — Friston & Kiebel (2009) <https://www.fil.ion.ucl.ac.uk/~karl/Predictive%20coding%20under%20the%20free-energy%20principle.pdf>
- **A Tutorial on the Free-Energy Framework for Modelling Perception and Learning** — Bogacz (2017) <https://www.sciencedirect.com/science/article/pii/S0022249615000759>
- **Predictive Coding: A Theoretical and Experimental Review** — Millidge et al. <https://arxiv.org/abs/2107.12979>

Predictive Coding Implementations (Yet to read)

- **An Approximation of the Backpropagation Algorithm in Predictive Coding Networks** — Whittington & Bogacz <https://www.sciencedirect.com/science/article/pii/S0893608017301032>
- **Predictive Coding Approximates Backprop Along Arbitrary Computation Graphs** — Millidge et al. <https://arxiv.org/abs/2006.04182>
- **Predictive Coding Networks and Inference Learning** <https://arxiv.org/abs/2407.04117>
- **Introduction to Predictive Coding Networks for Machine Learning** <https://arxiv.org/abs/2506.06332>

- **μPC: Scaling Predictive Coding to 100+ Layer Networks**
<https://arxiv.org/abs/2505.13124>
- **Towards the Training of Deeper Predictive Coding Neural Networks**
<https://arxiv.org/abs/2506.23800>
- **Towards Stable Learning in Predictive Coding Networks**
<https://openreview.net/forum?id=FwdN0KovFp>

Predictive Coding & World Models (Yet to read)

- **World Models and Predictive Coding for Cognitive and Developmental Robotics: Frontiers and Challenges** <https://arxiv.org/abs/2301.05832>
- **Learning Transformer-Based World Models with Contrastive Predictive Coding**
<https://arxiv.org/pdf/2503.04416>
- **Predictive World Models from Real-World Partial Observations**
<https://arxiv.org/abs/2301.04783>